

Alex Benson

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EDUCATION

Colorado State University

Expected Dec. 2027

Master of Science, Electrical Engineering

Fort Collins, CO

- Completed coursework: Electrical Power Engineering, Aerospace Vehicles Trajectory & Performance, Hardware/Software Design for Embedded Systems, AI for Radar & Remote Sensing
- Planned coursework: Embedded Systems & Machine learning, RF Integrated Circuit Design, Radar Systems & Design, Micro-Electro-Mechanical Devices.

Colorado State University

Expected May 2026

Bachelor of Science, Electrical Engineering; Concentration in Aerospace; Minor in Mathematics; 3.3/4.0

Fort Collins, CO

- Senator for the College of Engineering, Associated Students of Colorado State University (2022-2023)
- Chair for Engineers Without Borders: Raised \$7,000 for international aid projects (2022-2023)
- College of Engineering Dean's List (FA22, SP24)
- Leading payload development for CSU's NASA USLI competition rocket, managing design, testing, and integration.

WORK EXPERIENCE

FIRST RF

May 2025 – Aug. 2025

Embedded System Engineering Intern

Boulder, CO

- Co-developed a half-duplex phased array system for land and maritime platform connectivity.
- Designed and optimized multilayer PCB layouts in Altium, performing transmission-line calculations and stackup tuning to ensure signal integrity at GHz-range operating frequencies.
- Authored cross-platform SPI drivers in C for Zephyr RTOS, enabling control of active RF components.
- Automated analysis of passive component performance by processing S-parameter datasets in Python, accelerating validation workflows.

Bureau of Reclamation, U.S. Department of the Interior

May 2024 – Feb. 2025

Electrical Engineering Intern; Power System Analysis & Control Group

Lakewood, CO

- Performed hydropower governor and excitation control system compliance testing (WECC & NERC).
- Designed and implemented a field transducer box for signal processing at power plants across the west.
- Developed custom C drivers for interfacing with 12-/16-bit Acromag DAC PCIe boards in QNX RTOS for a governor and excitation system controller.
 - Integrated Simulink model for injection into DAC channels for dynamic simulation and testing.
 - Configured shared memory for Simulink model parameter adjustment via TCP/IP using Visual Basic.
- Developed 32-bit communication protocols for STM32 controller to interface with DAC peripheral board using SPI to upgrade prior 16-bit method.
 - Established frequency injection for a digital terminal quantity transducer, achieving a readable voltage range of -10 to +10 V to process 3-phase current and voltage from a hydropower turbine.

Electrical and Computer Engineering Department, Colorado State University

Aug. 2023 – May 2024

Learning Assistant

Fort Collins, CO

- Provided instruction, support, and tutoring for the Digital Circuit Logic course, including 5 additional office hours each week. Topics covered include combinational and sequential logic design, Boolean algebra, and basic circuit logic principles.

Woodward, Inc*Electrical Engineering Intern***May 2023 – Aug. 2023***Loves Park, IL*

- Researched and tested data acquisition hardware to improve the processes of the department and ultimately the testing done within the company.
- Designed and implemented a module to meet customer requirements on response tests for space hardware.
- Created a standalone Python script for rapid analysis of large data sets, enabling engineers to quickly visualize interactive graphs without need for NI DIAdem software.

Electronic-Photonic System Design Lab, Colorado State University*Undergraduate Researcher***Dec. 2022 – May 2023***Fort Collins, CO*

- Achieved a 75%-90% decrease in microchip measurement time, enhancing project turnaround and productivity.
 - Used Python libraries to isolate sectors of a microchip for the measurement process.

PROJECTS**NASA University Student Launch Initiative, Payload Team Lead****Fall 2025 – Spring 2026***Senior Design Project at Colorado State University*

- Leading a 6-member team to design and build a deployable soil collection and analysis payload for a 2026 launch.
- Collaborating with the vehicle team to ensure payload integration optimizes rocket performance and stability.
- Coordinating 5 separate 100+ page documents to support and justify the team's design decisions to NASA.
- All electrical designs, planning, and testing for the payload.

Arduino Monitored Ecosystem, environmental data collection system for plant growth.**Spring 2024**

- Utilized C++ to create a grab accurate environmental condition of an enclosed ecosystem.
- Enhanced this system on my own time to incorporate a feedback control system where environmental conditions would be adjusted if out of tolerance.
 - Foundational knowledge of control systems using relays.

Automated Greenhouse, robust greenhouse tent with controlled environmental parameters.**Fall 2024**

- This is a vertically integrated project (VIP) throughout the 2024 fall semester, requiring 1-3 hours/week.
- Took the original electrical design of the greenhouse control system and made it safe, efficient, and documented. Implementing the new system in the second iteration of the project.

ADDITIONAL SKILLS

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|----------|------------|----------------|--------------------|-------------------|
| ▪ Python | ▪ Assembly | ▪ Visual Basic | ▪ TCP/IP | ▪ PCB Design (RF) |
| ▪ Java | ▪ MATLAB | ▪ AutoCAD | ▪ Quartus Prime | ▪ Altium |
| ▪ C, C++ | ▪ LabVIEW | ▪ Cadence | ▪ HMI Dev | ▪ Proteus |
| ▪ RTOS | ▪ Simulink | ▪ Verilog | ▪ Embedded Systems | |